

1. Randomly set the center for K groups, let assume K = 2
2. Calculate the distance of each values to the center of each group, assign it to the group with least distance.
3. Recalculate the center of the group according to the result obtained in 2.
4. Repeat the process 2-3 until there is no change or where you are satisfied with the result.
5. Measure the performance using the total within-cluster sum of square (WSS)
6. Go to Step 1 for K = K+1, until K = maximum K
7. Select the best K using Knee-plot

Students	Physics	Biology
1	3.5	3
2	2	2.5
3	4	3.5
4	2	3
5	1	2
6	4	3

1 Center	Round 1	Physics	Biology	random students
	A	3.5	3	
	Round 1	B	4	random students
			3	

2 Distance
Euclidean

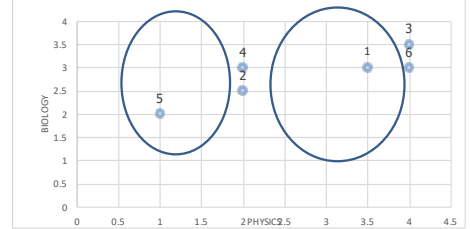
Students	Distance from A
1	0
2	1.58113883
3	0.707106781
4	1.5
5	2.692582404
6	0.5

Assigned to Cluster	สูตรคำนวณ
A	=IF(MIN(D25,D33)=D25,"A","B")
A	=IF(MIN(D26,D34)=D26,"A","B")
B	=IF(MIN(D27,D35)=D27,"A","B")
A	=IF(MIN(D28,D36)=D28,"A","B")
B	=IF(MIN(D29,D37)=D29,"A","B")
B	=IF(MIN(D30,D38)=D30,"A","B")

Students	Distance from B
1	0.5
2	2.061552813
3	0.5
4	2
5	3.16227766
6	0

5 Performance evaluation (the lower, the better)

	Sq Dist in A		Sq Dist in B	
1	0	=D25^2		
2	2.5	=D26^2		
3			0.25	=D35^2
4	2.25	=D28^2		
5	7.25	=D29^2		
6			0	=D38^2
RSS	12.25	=SUM(D43:F48)		
	RSS			
Round 1	12.25	=D49		
Round 2				



3 Group Re-center

Average ค่า Assigned to Cluster

Round 1 Group A	Physics	Biology
	2.125 =AVERAGE(C10,C11,C13,C14)	2.625 =AVERAGE(D10,D11,D13,D14)
Group B	4.000 =AVERAGE(C12,C15)	3.250 =AVERAGE(D12,D15)

Re Center	Round 2	Physics	Biology	random students
	A	2.125	2.625	
	Round 2	B	4.000	random students
			3.250	

Students	Distance from A
1	1.425219281
2	0.176776695
3	2.06911817
4	0.395284708
5	1.286953768
6	1.912132318

Assigned to Cluster	สูตรคำนวณ
B	=IF(MIN(D64,D72)=D64,"A","B")
A	=IF(MIN(D65,D73)=D65,"A","B")
B	=IF(MIN(D66,D74)=D66,"A","B")
A	=IF(MIN(D67,D75)=D67,"A","B")
A	=IF(MIN(D68,D76)=D68,"A","B")
B	=IF(MIN(D69,D77)=D69,"A","B")

Student Distance from B	
1	0.559016994
2	2.136000936
3	0.25
4	2.015564437
5	3.25
6	0.25

3 Group Re-center

Average ค่า Assigned to Cluster

Round 2 Group A	Physics	Biology
	1.667 =AVERAGE(C11,C13,C14)	2.500 =AVERAGE(D11,D13,D14)
Group B	3.833 =AVERAGE(C10,C12,C15)	3.167 =AVERAGE(D10,D12,D15)

	Sq Dist in A		Sq Dist in B	
1			0.3125	=D72^2
2	0.03125	=D65^2		
3			0.0625	=D74^2
4	0.15625	=D67^2		
5	1.65625	=D68^2		
6			0.0625	=D77^2
RSS	2.28125	=SUM(D82:F87)		
	RSS			
Round 1	12.25	=D52		
Round 2	2.28125	=D88		

Re Center	Round 3	Physics	Biology	random students
	A	1.667	2.500	
	Round 3	B	3.833	random students
			3.167	

Students	Distance from A
1	1.900292375
2	0.333333333
3	2.538591035
4	0.600925213
5	0.833333333
6	2.386303511

Assigned to Cluster	สูตรคำนวณ
B	=IF(MIN(D104,D112)=D104,"A","B")
A	=IF(MIN(D105,D113)=D105,"A","B")
B	=IF(MIN(D106,D114)=D106,"A","B")
A	=IF(MIN(D107,D115)=D107,"A","B")
A	=IF(MIN(D108,D116)=D108,"A","B")
B	=IF(MIN(D109,D117)=D109,"A","B")

Students	Distance from B
1	0.372677996
2	1.950783318
3	0.372677996
4	1.840893503
5	3.064129385
6	0.23570226

3 Group Re-center

Average ค่า Assigned to Cluster

Round 3 Group A	Physics	Biology
	1.667 =AVERAGE(C11,C13,C14)	2.500 =AVERAGE(D11,D13,D14)
Group B	3.833 =AVERAGE(C10,C12,C15)	3.167 =AVERAGE(D10,D12,D15)

	A	B	C	D	E	F	G	H	I	J	K	L	M
121				Sq Dist in A			Sq Dist in B						
122			1				0.138888889						
123			2	0.111111111	=D105^2								
124			3				0.138888889						
125			4	0.361111111	=D107^2								
126			5	0.694444444	=D108^2								
127			6				0.055555556						
128			RSS	1.5	=SUM(D122:F127)								
129													
130				RSS									
131			Round 1	12.25	=D52								
132			Round 2	2.28125	=D92								
133			Round 3	1.5	=D128								
134													
135													
136													
137													
138													

สรุป เมื่อ สมาชิก Assigned to Cluster(center) ไม่เปลี่ยนไปจากรอบก่อนหน้าให้หยุดทำแล้วสรุปได้ดังนี้
center ของgroupA ที่ดีที่สุดคือ 1.667 และ 2500
center ของgroupB ที่ดีที่สุดคือ 3.833 และ 3.167